

DEERING, AK, USA

WMO: 701333

Lat: 66.069N

Lon: 162.764W

Elev: 21

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 26643

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.9	(c) -30.1	(d) -33.3	(e) 0.8	(f) -27.7	(g) -31.1	(h) 1.0	(i) -25.5	(j) 35.8	(k) 14.4	(l) 32.7	(m) 11.9	(n) 4.9	(o) 220	(p) 0.908

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	68.5	58.6	65.2	57.1	62.5	55.4	59.9	66.5	57.9	63.8	56.1	61.6	10.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.6	68.2	63.4	54.5	63.3	61.3	52.7	59.2	59.5	26.4	66.5	25.0	63.8	23.9	61.5	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 30.7	(b) 26.8	(c) 24.2	(e) -38.0	(f) 75.6	(g) 4.4	(h) 5.6	(i) -41.1	(j) 79.6	(k) -43.7	(l) 82.9	(m) -46.2	(n) 86.0	(o) -49.4	(p) 90.1		
(4) 30.7	(5) 26.8	(6) 24.2	(7) -31.7	(8) 61.4	(9) 1.8	(10) 2.4	(11) -33.0	(12) 63.1	(13) -34.0	(14) 64.4	(15) -35.1	(16) 65.8	(17) -36.4	(18) 67.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	-1.9	2.5	1.2	14.5	32.9	46.9	53.5	50.2	42.4	28.8	13.1	3.6
	DBStd	23.29	17.91	17.69	14.06	12.44	9.54	6.78	5.06	5.39	6.22	9.11	12.55	15.92
	HDD50	9712	1608	1329	1514	1065	531	137	25	66	235	658	1106	1438
	HDD65	14931	2073	1749	1979	1515	995	543	358	460	677	1123	1556	1903
	CDD50	258	0	0	0	0	2	43	132	72	9	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	17	0	0	0	0	0	1	13	3	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
	WSAvg	10.5	11.3	12.9	11.3	10.9	9.9	9.2	9.1	9.7	10.3	10.2	10.4	11.3
	PrecAvg	12.2	0.7	0.9	0.5	0.6	0.5	0.6	1.6	2.3	1.6	1.2	0.9	0.8
(15) Precipitation	PrecMax	17.2	1.4	4.3	2.2	1.9	1.5	1.8	3.7	5.5	2.9	3.4	2.3	2.4
	PrecMin	8.3	0.0	0.0	0.1	0.1	0.1	0.0	0.4	0.1	0.4	0.2	0.0	0.3
	PrecStd	2.4	0.4	0.9	0.4	0.4	0.4	0.4	0.8	1.3	0.7	0.7	0.6	0.5
	PrecStd	2.4	0.4	0.9	0.4	0.4	0.4	0.4	0.8	1.3	0.7	0.7	0.6	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	45.4	37.1	43.4	60.9	71.9	74.8	70.1	61.0	49.0	38.8	37.9
		MCWB	35.1	38.2	35.1	38.7	49.2	57.8	61.7	58.5	52.6	44.3	35.8	35.0
	2%	DB	34.0	35.7	33.0	37.7	54.3	66.4	69.5	65.0	57.1	45.5	36.1	33.8
		MCWB	31.5	32.4	30.9	34.6	46.6	55.9	59.7	57.7	51.4	42.4	33.8	31.5
	5%	DB	29.2	32.4	29.2	35.6	49.8	62.8	66.7	62.4	54.7	43.1	33.9	30.4
		MCWB	27.1	29.8	27.0	33.1	44.0	54.0	58.5	56.6	49.9	40.5	31.8	28.1
	10%	DB	24.7	27.5	25.1	32.8	46.0	59.2	63.7	59.8	52.3	40.5	30.4	26.1
		MCWB	22.9	25.2	23.4	30.7	41.2	51.7	56.5	54.9	48.3	38.2	28.8	24.1
	0.4%	WB	35.1	38.9	35.3	38.9	49.9	60.3	63.3	60.5	54.2	45.3	36.0	35.2
		MCDB	38.2	46.2	37.5	43.3	58.6	67.1	70.7	66.4	57.9	48.5	38.1	37.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.9	32.6	31.1	35.0	47.0	56.8	61.2	58.8	52.2	42.6	33.8	31.6
		MCDB	34.2	35.4	33.0	37.7	54.1	65.1	68.5	63.8	56.0	45.0	35.6	33.4
	5%	WB	27.7	28.4	26.1	33.1	44.2	54.4	59.1	57.1	50.4	40.7	31.8	28.0
		MCDB	29.7	30.7	28.1	35.4	49.4	61.7	65.9	61.6	54.3	43.1	33.6	30.1
	10%	WB	23.4	23.8	20.8	30.9	41.3	52.0	57.0	55.3	48.5	38.4	29.1	24.1
		MCDB	25.2	26.1	22.3	32.9	45.5	58.5	63.0	59.6	52.0	40.5	30.4	25.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	14.5	14.3	13.2	12.7	15.8	14.6	13.5	12.7	10.7	13.4	13.7
		MCDBR	14.9	15.5	14.5	13.5	19.8	21.9	22.3	17.6	15.3	9.8	10.8	13.6
		MCWBR	13.6	14.1	15.3	11.3	12.9	12.5	13.1	11.5	10.5	7.9	9.7	12.2
	5% WB	MCDBR	14.8	16.2	16.2	12.8	19.3	20.2	20.8	16.0	14.8	9.4	9.7	13.4
		MCWBR	13.5	14.1	15.3	10.9	12.8	12.1	13.0	11.0	10.6	7.7	9.0	12.4
(40) Clear-Sky Solar Irradiance	taub		0.175	0.242	0.262	0.328	0.353	0.346	0.359	0.373	0.305	0.262	N/A	0.172
	taud		1.900	2.132	2.218	2.062	2.136	2.362	2.419	2.391	2.531	2.356	N/A	1.918
	Ebn at Noon		86	200	258	260	266	273	266	247	246	196	N/A	13
	Edh at Noon		6	17	25	39	40	33	31	29	20	13	N/A	1
	RadAvg		31	220	784	1446	1805	1928	1500	1097	683	247	55	4
(45) All-Sky Solar Radiation	RadStd		3	26	62	89	182	141	123	130	67	28	8	0

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2

Nomenclature: See separate page